CityUniversity of Seattle

School of Technology and Computing

From Anxiety to Confidence

A realistic, research-backed guide for career changers starting a tech degree

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Why This Guide Exists

If you're starting a technology program without a technical background, the mix of excitement and dread you're feeling is normal — and it's shared by a large and growing group of people. Tech is not a closed club anymore, and the data backs that up.

> *The U.S. tech occupation workforce reached roughly 5.9 million people in 2024 and was projected to grow to about 6.1 million in 2025 — a field expanding fast enough to need people from every kind of background. (CompTIA, State of the Tech Workforce 2025)*
>
> *Nearly half of all U.S. tech job postings in June 2025 did not require a four-year degree. (CompTIA, Tech Jobs Report, July 2025)*

This guide walks through what the anxiety actually is, why it doesn't mean you don't belong, and what specifically works — grounded in research, not just encouragement — to build real competence and confidence over the course of your program.

1. The Imposter Syndrome Myth

Imposter syndrome — the feeling that you don't really belong or aren't smart enough, despite evidence to the contrary — is one of the most common experiences reported by people entering or re-entering tech. It is not a sign you've made a mistake; it's close to the norm.

> *In a 2023 survey of 250 women trying to re-enter the technology industry, nearly all identified imposter syndrome and low confidence as the single biggest barrier to returning, and 78% said they wanted more mentoring and wellbeing support from employers and training programs. (Tech Returners, 2023, reported via IT Pro)*
>
> *A KPMG study of 750 high-performing women executives found 75% had experienced imposter syndrome at points in their careers, and 47% linked the feeling specifically to never expecting to reach the level of success they'd already achieved. (KPMG, "Mind the Gap" study, 2020)*

What this looks like in practice

The myths worth naming directly: that everyone else is naturally good at this (technical skill is learned, not innate); that you're too old or too late (career changers regularly start in their 30s, 40s, and beyond); that you lack a "programmer brain" (if you can debug why your Wi-Fi is down, you already use the core skill); and that real developers don't look things up (professional developers spend a large share of their working day searching documentation and references).

- Keep a running list of moments you solved something without help — review it when self-doubt spikes.
- Notice when you're comparing your first attempt to someone else's hundredth.
- Say the myth out loud to a classmate or instructor — naming it out loud usually deflates it.

2. You Are Not Alone: The Rise of the Adult Tech Learner

Non-traditional students returning to school for technology are not a small niche — they are one of the fastest-growing groups in U.S. higher education, and colleges are increasingly built around exactly this kind of student.

> *Adults age 25 and older make up roughly a third of all U.S. postsecondary enrollment, and first-year enrollment growth among students 25 and older has recently been outpacing growth among traditional-age students. (Jobs for the Future, "Adult Learners: A Literature Review," February 2025, citing NCES data)*

What this looks like in practice

Your non-technical background is also a functional advantage, not just a starting-point disadvantage. People coming from other fields bring communication skills, a user's-eye view of what technology needs to actually do, domain expertise the industry doesn't have enough of, and the resilience of having already succeeded in one hard field before.

- Write down two skills from your previous field or job that transfer directly (project management, client communication, data analysis, teaching, etc.).
- Find one other adult learner in your cohort in the first two weeks — even a single peer changes the experience of a hard term.
- Ask your advisor what support services exist specifically for returning or working-adult students; most schools built for this population have them.

3. You Don't Need a Computer Science Degree to Start

The assumption that everyone in tech followed a straight line from a CS degree to a coding job is false for the majority of working developers today, and employers' own hiring practices increasingly reflect that.

> *66% of professional developers hold a bachelor's or master's degree, yet only 49% say they actually learned to code in school — most taught themselves or used other resources. 82% cite online resources as a top way they learn to code. (Stack Overflow, 2024 Developer Survey)*

What this looks like in practice

Because the traditional path isn't the dominant path, the practical foundation that actually matters is: problem decomposition (breaking a big problem into small, testable pieces), reading documentation and error messages closely, and building a small portfolio of real, working projects rather than memorizing terminology.

- Keep a personal glossary of terms as you meet them — understand the concept before worrying about the vocabulary.
- Use official documentation (MDN for web, language docs for your stack) as your first stop, the way most working developers do.
- Commit your first small project to GitHub in week one, even if it's tiny — a public record of progress builds real evidence of capability.

4. Learning Techniques That Actually Work

Some study techniques are meaningfully better than others for technical material, and federally funded classroom research backs a specific set of them.

> *A federally funded meta-analysis of classroom interventions found that structured, active-learning approaches and near-peer support measurably improve STEM students' motivation and persistence. (National Science Foundation, Award #2110368, "Improving Undergraduates' Motivation and Retention in STEM Through Classroom Interventions: A Meta-Analysis")*

What this looks like in practice

The Feynman technique — explaining a concept in plain language until the gaps in your own understanding show up — works because it forces active recall rather than passive review. The same is true of writing code by hand instead of copy-pasting, intentionally breaking working code to see what fails, and spacing review sessions out over days rather than cramming once.

- Use the 5-minute rule when stuck: 5 minutes solving alone, 5 minutes researching, then ask for help — this builds independence without letting frustration spiral.
- Review new material at increasing intervals (next day, then a week later, then a month later) instead of once.
- Build one small real project per module — a script, a page, a tool that solves something you actually have.

5. Building Your Support Network

The single biggest reported barrier for adults entering or returning to tech is confidence, and the intervention with the clearest evidence behind it is mentorship and peer support — which is exactly what the returners in the Tech Returners survey said they wanted more of.

What this looks like in practice

Support doesn't have to be formal to work. A consistent study group, regular office-hours attendance, and one person you can ask a specific, well-described question of when you're stuck for more than an hour or two all measurably reduce the odds of quietly disengaging from a hard program.

- Join or start a study group that meets on a fixed schedule, not an ad hoc one.
- Attend office hours at least once in the first month, even with a small question — it normalizes asking.
- Look for a program mentor or an upperclassman willing to talk through what the first term actually felt like for them.

How CityU's School of Technology and Computing Builds These Skills

CityU of Seattle's School of Technology and Computing is built for working adults and career changers — every class is offered online, and the program ladder is designed to let you start small and build up rather than requiring a running start.

- Undergraduate Certificate in Foundations of Computing — a low-stakes, structured way to test the field before committing to a full degree.
- Undergraduate Certificate in Fundamentals of Computing — builds core technical literacy for students without a prior technical background.
- Bachelor of Science in Applied Computing — a foundational degree covering programming, web development, IT, and systems analysis, designed with working adults in mind.

- Master of Science in Computer Science — for career changers who complete a foundation and want to go further into the field professionally.

Your Next Step

You don't need to resolve every doubt before you start — you need one small, concrete next step. Pick one: talk to a CityU advisor about which certificate or degree fits where you are right now, join a study group or find one peer in your first two weeks, or commit your first small project to GitHub this month. Progress in a program like this is rarely about speed. It's about staying in motion long enough for the small wins to add up — and CityU's advisors are there to help you figure out what that first step looks like.

Sources

- CompTIA, "State of the Tech Workforce 2025"
- CompTIA, "Tech Hiring Activity Outpaces Expectations" (Tech Jobs Report, July 2025)
- Stack Overflow, 2024 Developer Survey — Developer Profile
- Jobs for the Future (JFF), "Adult Learners: A Literature Review" (February 2025)
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